Operating Instructions

These are the operating instructions for the DIY LED Clock firmware found here:
https://github.com/fabianswebworld/stc-led-clock-firmware

These instructions apply to firmware release: **1.3.0**

1. Buttons and indicators

The clock is operated using two push-buttons located on the side or rear of the unit:

- **Button 1 (S1, top):** Settings menu / Advance menu item / Select value
- **Button 2 (S2, bottom):** Next display page / Select menu item / Increment value / Functions menu

The dots on the display are used as indicators:

- **Center Colon (`:`):** Flashes once per second with an incandescent filament effect (gradual ignition and afterglow decay).
- **Indicator Dot:**
  - **In 24-hour display mode:** The decimal point at the lower-right corner of the fourth digit stays steadily lit when the daily alarm is armed.
  - **In 12-hour display mode:** The decimal point at the lower-right corner of the fourth digit indicates PM.

In normal operation, the clock displays the current hours and minutes. Additionally, at configurable intervals, an "auto-rotation" through other items (Temperature, Date, Day of Week) is performed. For each of these three items (Temperature, Date, Day of Week), the inclusion in the auto-rotation can be enabled or disabled.

2. Actions from normal time display

- **Button 1 (Short press):** Enters the Settings menu (see Section 7).
- **Button 1 (Hold for 1 second):** Launches the 7-segment Snake animation manually.
- **Button 2 (Short press):** Steps to the next display page (see Section 3).
- **Button 2 (Hold for 1 second):** Enters the Functions Menu (see Section 4).

3. Manual display of other information

Pressing **Button 2** steps through the available information displays in sequence:

Time → Temperature → Date → Day of Week → Year → Seconds → Time

- **Temperature:** Displays ambient temperature (e.g., `22°C` or `72°F`).
- **Date:** Displays month and day according to the selected format (e.g., `31.12.`, `31.12`, or `12.31`).
- **Day of Week:** Displays three-letter abbreviation of the current weekday.
- **Year Display (`'xx`):** Displays the last two digits of the year preceded by an apostrophe at the top (e.g., `'26` for 2026). The internal clock circuit automatically calculates leap years.
- **Seconds Display (`MM.SS`):** Displays elapsed minutes on the left digits and running seconds on the right digits.
  - **Time Synchronization:** Pressing **Button 1** while on this page immediately resets the seconds counter to `.00` to synchronize with a reference clock.

When manually advancing the modes, all of the above pages are displayed, regardless of whether they are also enabled for the auto-rotation in the `dISP` menu (see Section 5). A page selected manually with **Button 2** stays on the screen indefinitely, the clock will not switch back automatically.

- **Returning to Clock:** Pressing **Button 2** repeatedly until the hours and minutes return restores normal timekeeping and re-enables automatic rotation.

4. Countdown Timer (Ctdn) and Stopwatch (Stop)

A countdown timer and a stopwatch are available from the Functions Menu.

4.1 Accessing the Functions Menu

1. **Activation:** While in normal clock mode, **press and hold Button 2**. The display shows Ctdn.
2. **Menu Navigation:**
    - **Button 1:** Cycles through the available functions:
      Ctdn → Stop → Back to normal clock display.
    - **Button 2:** Confirms and enters the selected function. From any of the functions, you can **hold Button 2** to return to the normal clock display.

4.2 Countdown Timer (Ctdn)

1. **Setting the Duration (MM:SS):**
    - The display shows the last used timer preset (factory default: 00:59).
      The colon remains steady.
    - **Button 1:** Increments minutes (from 00 up to 99).
    - **Button 2:** Increments seconds (from 00 up to 59).
    - **Button 1 + Button 2 (simultaneously):** Resets the timer to 00:00.
    - **Hold Button 2:** Cancels the countdown and returns to normal clock display.
2. **Automatic Start:**
    - After not pressing any button for **2 seconds**, the countdown begins automatically (the chosen starting value is saved to memory as the new default preset).
    - The colon flashes while counting down.
3. **While Running:**
    - Pressing any button while running immediately pauses the countdown and returns to the Duration setting, allowing you to add more time, starting from the current value.
    - **Hold Button 2:** Cancels the countdown and returns to normal clock display.
4. **Timer Alarm:**
    - When reaching 00:00, the display flashes rapidly and the buzzer sounds.
    - **Pressing either Button 1 or Button 2** silences the alarm and immediately returns to the normal clock display.

4.3 Stopwatch (Stop)

1. **Display Format (SS.hh):**
    - Displays seconds and hundredths of a second (00.00 to 99.99).
    - The stopwatch always initializes to 00.00 (paused) upon entry, with the top indicator dot illuminated steady.
2. **Controls:**
    - **Button 1:** Starts or pauses the stopwatch.
    - **Button 2 (while paused):** Resets the display back to 00.00.
    - **Hold Button 2:** Exits the stopwatch and returns to normal clock display.
3. **While Running:**
    - The hundredths digits run continuously at high speed, and the top indicator dot blinks.
    - You can pause and resume timing as often as needed.
    - Upon reaching 99.99 seconds, the stopwatch automatically rolls over to 00.00 and continues counting uninterrupted.

5. Automatic Display Rotation

If enabled in the `diSP` menu (see Section 7), the clock will smoothly dissolve from the time display through the enabled information pages (Temperature, Date, Day of Week) before returning to the clock. This rotation can be configured in the `dISP` menu (see Section 7):

- **Cadence/Interval Options (`rot`):**
  - `rot0`: Automatic rotation disabled (manual cycling only).
  - `rot1`: Once per minute (at second :30).
  - `rot2`: Twice per minute (at seconds :00 and :30).
  - `rot3`: Three times per minute (at seconds :00, :20, and :40).
- **Page Duration (`dur`):**
  - Configurable from `dur1` **to** `dur5` (each page holds for 1 to 5 seconds).

6. Hourly animation (`Ani`)

A fun little snake animation starts: the snake slithers across the display segments, locates "apples", grows in length, and dissolves back into the clock display. The animation lasts for approximately 16 seconds and can be cancelled at any time by pushing one of the buttons, exiting back to normal time display.

- **Cadence Options (`Ani`):**
  - `Ani0`: Animation disabled.
  - `Ani1`: Twice daily (at 12:00 noon and 00:00 midnight).
  - `Ani2`: Once every hour (at the top of the hour, :00), default.
  - `Ani3`: Every half hour (:00 and :30).
  - `Ani4`: Every quarter hour (:00, :15, :30, :45).
  - `Ani5`: Every minute (at second :00).
- **Manual Trigger:** Holding **Button 1** for 1 second in normal clock mode launches the animation immediately.

7. Settings Menu

Press **Button 1** from normal clock mode to enter the Settings menu tree.

- Use **Button 1** to cycle through the four top-level menus (`SEt`, `bEEP`, `diSP`, `ConF`).
- Use **Button 2** to enter a menu.
- Inside any setting: **Button 2** changes the value, **Button 1** confirms and advances to the next setting.

```
[Normal Clock]
     |
  (Button 1)
     |
     +---> [ SEt  ] ---> Clock Time -> Alarm Time -> Chime Hours (begin/end) -> Date -> Day of Week
     |                   -> Year
     |
     +---> [ bEEP ] ---> Alarm Toggle -> Hourly Chime Toggle
     |
     +---> [ diSP ] ---> Temperature Enable -> Date Enable -> Day of Week Enable -> Rotation cadence
     |                   -> Page duration -> Animation cadence
     |
     +---> [ ConF ] ---> 12/24h -> Leading Zero -> Temperature Unit -> Date Format -> Brightness
                         -> Temperature Calibration -> Firmware Version
```

Menu 1: Set Clock, Alarm, Date (`SEt`)

Press **Button 2** on `SEt` to enter the setting sequence:

1. **Clock Time (`CL`):** Hours flash → press **Button 2** to adjust → press **Button 1** → Minutes flash → press **Button 2** to adjust → press **Button 1**.

2. **Daily Alarm Time (**AL**):** Alarm hours flash → adjust with **Button 2** → press **Button 1** → Alarm minutes flash → adjust with **Button 2** → press **Button 1**.
3. **Hourly Chime Window (**CH**):** Sets active hours for the chime.
    - Start hour flashes → adjust with **Button 2** → press **Button 1**.
    - End hour flashes → adjust with **Button 2** → press **Button 1**.
4. **Date (**DA**):** Month flashes → adjust with **Button 2** → press **Button 1** → Day flashes → adjust with **Button 2** → press **Button 1**.
5. **Day of Week (**DY**):** Day flashes → press **Button 2** to adjust (Sun ... Mon) → press **Button 1**.
6. **Year (**Yr**):** Year flashes → press **Button 2** to adjust (00 ... 99) → press **Button 1** to save and exit to clock.

Menu 2: Sound Settings (bEEP)

Press **Button 2** on bEEP to enter:

1. **Alarm Enable:** Displays AL on or AL oF. Press **Button 2** to toggle. Press **Button 1** to confirm.
2. **Chime Enable:** Displays CH on or CH oF. Press **Button 2** to toggle. Press **Button 1** to confirm and exit to clock.

Menu 3: Display Settings (diSP)

Press **Button 2** on diSP to enter settings for the display rotation:

1. **Temperature Page:** Displays tP on or tP oF. Press **Button 2** to toggle. Press **Button 1** to advance.
2. **Date Page:** Displays dA on or dA oF. Press **Button 2** to toggle. Press **Button 1** to advance.
3. **Day of Week Page:** Displays dy on or dy oF. Press **Button 2** to toggle. Press **Button 1** to advance.
4. **Auto-Rotation Cadence (**rot**):** Displays rot0, rot1, rot2, or rot3 (see Section 5). Press **Button 2** to cycle. Press **Button 1** to advance.
5. **Page Duration (**dur**):** Displays dur1 to dur5 (seconds to show each page in rotation). Press **Button 2** to cycle. Press **Button 1** to advance.
6. **Animation Cadence (**Ani**):** Displays Ani0 to Ani5 (see Section 6). Press **Button 2** to cycle. Press **Button 1** to save and exit to clock.

Menu 4: Configuration (ConF)

Press **Button 2** on ConF to enter configuration settings:

1. **Time Format (**12.24**):** Press **Button 2** to toggle between 12 (12-hour AM/PM) and 24 (24-hour). Press **Button 1** to advance.
2. **Leading Zero Suppression (**ZEro**):** Press **Button 2** to toggle between 09:59 (leading zero visible) and 9:59 (leading zero blanked). Press **Button 1** to confirm and advance.
3. **Temperature Unit (**Unit**):** Press **Button 2** to toggle between °C (Celsius) and °F (Fahrenheit). Press **Button 1** to advance.
4. **Date Format (**DATE**):** Press **Button 2** to cycle through:
    - 12.31 (Month.Day)
    - 31.12 (Day.Month)
    - 31.12. (Day.Month with trailing point, German format)
    - Press **Button 1** to confirm and advance.
5. **Automatic Dimming Range (**brt**):**
    - Press **Button 2** to set maximum brightness level (01 ... 63).
    - Press **Button 1**, then **Button 2** to set minimum (night-time) brightness level (01 ... 63).
    - Press **Button 1** to advance.
6. **Temperature Sensor Calibration (**CALI**):**
    - Press **Button 2** to enter → current temperature reading is displayed.
    - Press **Button 2** to increment offset (-15 ... +15 degrees).

    - Press **Button 1** to advance.
7. **Firmware Version Display (`vEr`):**
    - Press **Button 2** to enter -> display shows installed firmware revision (e.g. `1.3.0`).
    - Press **Button 1** to exit to clock.

8. Backup Battery

The clock uses an onboard CR1220 3V coin cell battery to back up the DS1302 real-time clock circuit.

- **Retained Data:** Timekeeping, leap-year calculations, alarm settings, rotation preferences, countdown presets, and sensor calibrations are retained in non-volatile battery-backed memory.
- **Mains Reconnection:** When main power is disconnected and restored, the display immediately resumes operation with current time and customized settings intact without requiring reconfiguration.

9. Factory reset

To reset the clock to its factory defaults or to initialize in case of corrupt data in the RTC RAM, please follow these instructions:

- Remove power
- Press and hold **both buttons** (S1 and S2)
- While holding the buttons, reapply power
- After 1 second, **release the buttons**. The display will go blank and as soon as both buttons are released, display the Snake Animation once (~ 16 s) to confirm reset. To skip the animation, press any button during the animation, which will exit to normal clock display (0:00).
- The clock defaults to the following settings: 24-hour, DD.MM. format, leading zero suppression on, temperature unit °C, automatic display of temperature and date enabled, Snake animation enabled at top of hour.

Fabian Schneider (@fabianswebworld), last modified: 2026-10-16